

# TrueNAS + WD: Powering Enterprise Data Infrastructure at Scale



## Challenges

- Supporting continued capacity growth without increasing operational risk
- Ensuring predictable performance and reliability over extended deployment lifecycles
- Maintaining high availability through dual controller SAS architectures
- Qualifying new drive and platform technologies early—without sacrificing reliability or efficiency

## Solution

TrueNAS leveraged WD's Ultrastar® Series Hybrid Storage Platforms and 26TB<sup>1</sup> Ultrastar DC HC670 SAS HDDs to deliver high availability and enterprise-grade reliability. ArcticFlow™ cooling, IsoVibe™ vibration isolation, and helium-sealed drives enable dense, power-efficient storage with consistent performance at scale.

## Key Outcomes

- TrueNAS qualified high-capacity WD hard drives and WD storage platforms for high availability deployments
- Enabled long lifecycle customer environments supported by proven WD reliability
- Positioned the latest TrueNAS V Series to scale up to 30PB
- Delivered consistent performance, thermal stability, and platform reliability in mission critical use cases

## Background

TrueNAS delivers high-performance, highly reliable enterprise storage systems designed for mission-critical environments and long deployment lifecycles. Often deployed for AI/HPC, virtualization, backup, media & entertainment, CCTV/surveillance, and large data repositories, TrueNAS® Enterprise supports predictable file, block, and object (S3) workloads.

Since 2014, TrueNAS has partnered with WD, qualifying and shipping WD HDDs across five generations of TrueNAS platforms. TrueNAS Enterprise can scale to tens of petabytes using WD storage platforms, with proven deployments in commercial, research, government, and defense applications.

## Scaling Capacity Without Compromising Reliability

As enterprise customers increasingly prioritize longer lifecycle deployments, higher capacity per rack, and uncompromising reliability, TrueNAS required storage platforms and drives that could consistently perform in mission critical, high availability environments.

TrueNAS needed to scale capacity to tens of petabytes while maintaining the reliability, thermal stability, and availability required for enterprise and mission critical deployments over many years of operation. Dense storage configurations and long lifecycle expectations placed increased importance on consistent performance, effective thermal management, and vibration control.

## Enterprise Platforms Qualified for High Availability

TrueNAS standardized on WD's Ultrastar Data60 and Data102 platforms for its high capacity expansion shelf offerings. It validated the platforms through extensive qualification and real-world deployment. WD's consistent engineering collaboration and platform level design has enabled TrueNAS to confidently support enterprise customers with predictable performance and reliability over extended lifecycles.

The dual controller SAS architecture plays a critical role in maintaining high availability, ensuring uninterrupted access for business-critical workloads. WD's reliability extends beyond the drives themselves, encompassing the full platform—from chassis design and airflow optimization to vibration mitigation—making it well suited for TrueNAS' mission critical deployments.



Ultrastar Data102 Hybrid Storage Platform

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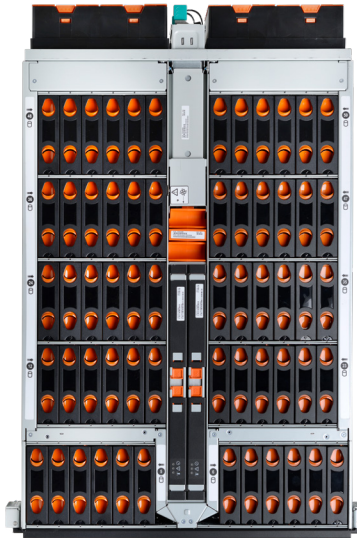
## Delivering Results Across Generations of Deployment

With WD platforms and HDDs at the foundation of its enterprise offerings, TrueNAS has:

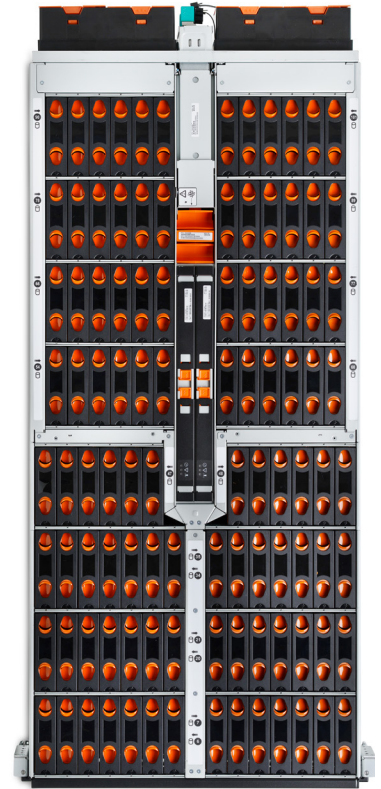
- Delivered five generations of enterprise storage products backed by WD HDDs
- Supported customers seeking longer lifecycles without compromising availability
- Benefited from WD's early, reliable adoption of new drive technologies, including helium-based designs for lower power profiles
- Maintained consistent system reliability and performance across dense, high-capacity configurations

## A Proven Partnership for Enterprise Data Infrastructure

As enterprise workloads become more data-intensive, this proven foundation gives customers a scalable, reliable, and open path for future storage, analytics, and AI infrastructure needs. The long-standing partnership between TrueNAS and WD demonstrates how reliable platforms and collaborative engineering enable enterprise storage systems built for longevity, efficiency, and scale. By leveraging WD technologies, TrueNAS continues to deliver trusted, high availability solutions that meet the evolving needs of enterprise customers—while positioning TrueNAS Enterprise platforms to scale confidently to 30PB and beyond.



Ultrastar Data60 Hybrid Storage Platform



Ultrastar Data102 Hybrid Storage Platform



<sup>1</sup>One terabyte (TB) is equal to one trillion bytes and one petabyte (PB) is equal to 1,000 TB. Actual user capacity may be less due to operating environment.